Topic: Small Enterprise Network.

This project focuses on designing and implementing a network that fulfills the company's operational and technical requirements.

OBJECTIVES

- To design a network using CISCO's Network devices.
- To segment departments using VLANs.
- To provide wireless connectivity to all departments.
- To enable automatic IP address assignment using DHCP.
- To ensure communication between all departments via inter-VLAN routing.
- To maintain scalability and security.

NETWORK EQUIREMENTS.

Department Structure:

The Company is consists of Three Departments

1. Admin
2. HR
3. Finance

Hardware Requirements

1. 1 Router (2911)
2. 1 Switch
3. End Devices (PCs, Printers, Access Points, Smart Phone, Laptop)

Functional Requirements

- Each department is in Separate VLAN
- Each department has Wireless Access.
- Devices receive IP addresses automatically
- Inter- Department communication must be enabled.

NETWORK DESIGN

Base Network: 192.168.1.0/24

Department	VLAN ID	Subnet	Gateway
ADMIN	10	192.168.1.0/26	192.168.1.1
HR	20	192.168.1.64/26	192.168.1.65
FINANCE	30	192.168.1.128/26	192.168.1.129

VLAN CONFIGURATION

Each department is isolated using VLANs:

- VLAN 10 -> Admin
- VLAN 10 -> HR
- VLAN 10 -> Finance

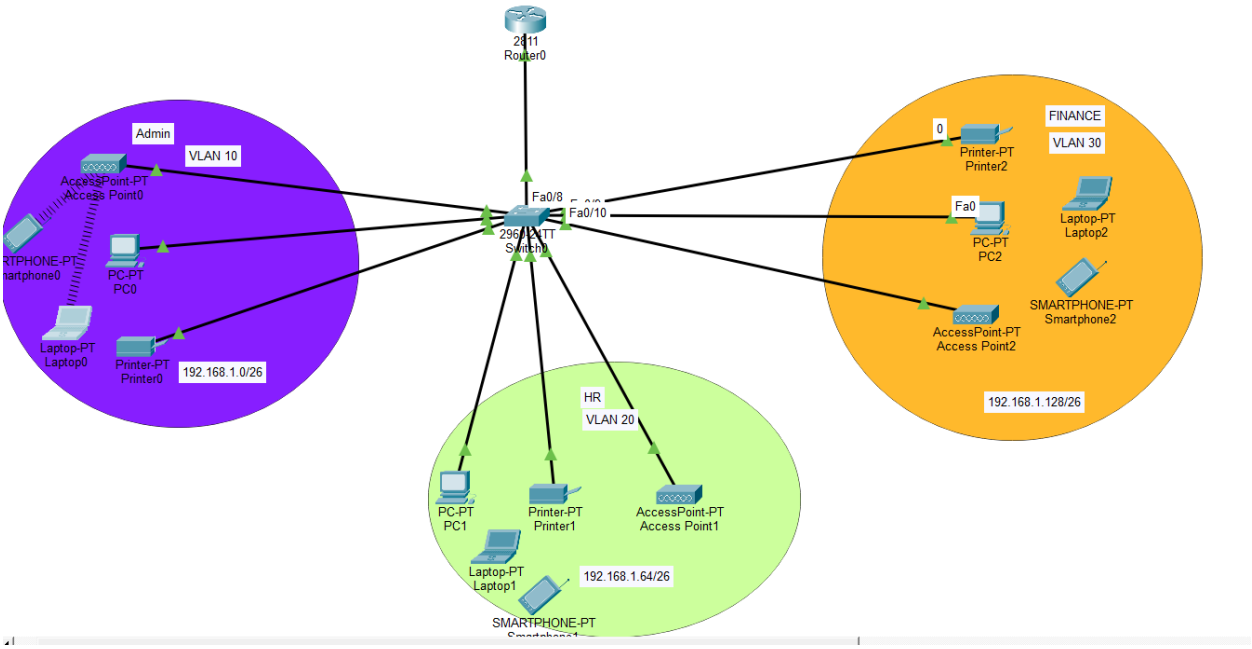
This Improves:

- Security.
- Traffic Management.
- Network Organization.

NETWORK TOPOLOGY

Topology Type: Star Topology

- Router Connected to Switch.
- Switch connects to all devices.
- Wireless Access Points connected to switch.
- Each access point mapped to respective VLAN



INTER-VLAN ROUTING

To allow communication between departments:

Configure sub-interfaces

- Fa0/0.10 -> VLAN 10
- Fa0/0.20 -> VLAN 20
- Fa0/0.30 -> VLAN 30

DHCP CONFIGURATION

Router will act as DHCP Server:

- VLAN 10: 192.168.1.2 – 192.168.1.62
- VLAN 20: 192.168.1.66 – 192.168.1.126
- VLAN 30: 192.168.1.130 – 192.168.1.190

WIRELESS NETWORK DESIGN

Each department will have its own Wi-Fi:

Department	SSID Name	VLAN
Admin	Admin-WiFi	10
HR	HR-WiFi	20
Finance	Finance – WiFi	30

- Secure WPA/2
- Connected through access points

CONFIGURATION OF VLANS on SWITCH

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport mode trunk
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/3
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/4
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/7
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/8
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/9
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/10
 switchport access vlan 30
 switchport mode access
```

ROUTER CONFIGURATION

```
 speed auto
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.1.1 255.255.255.192
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.1.65 255.255.255.192
!
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.1.129 255.255.255.192
```

DHCP CONFIGURATION

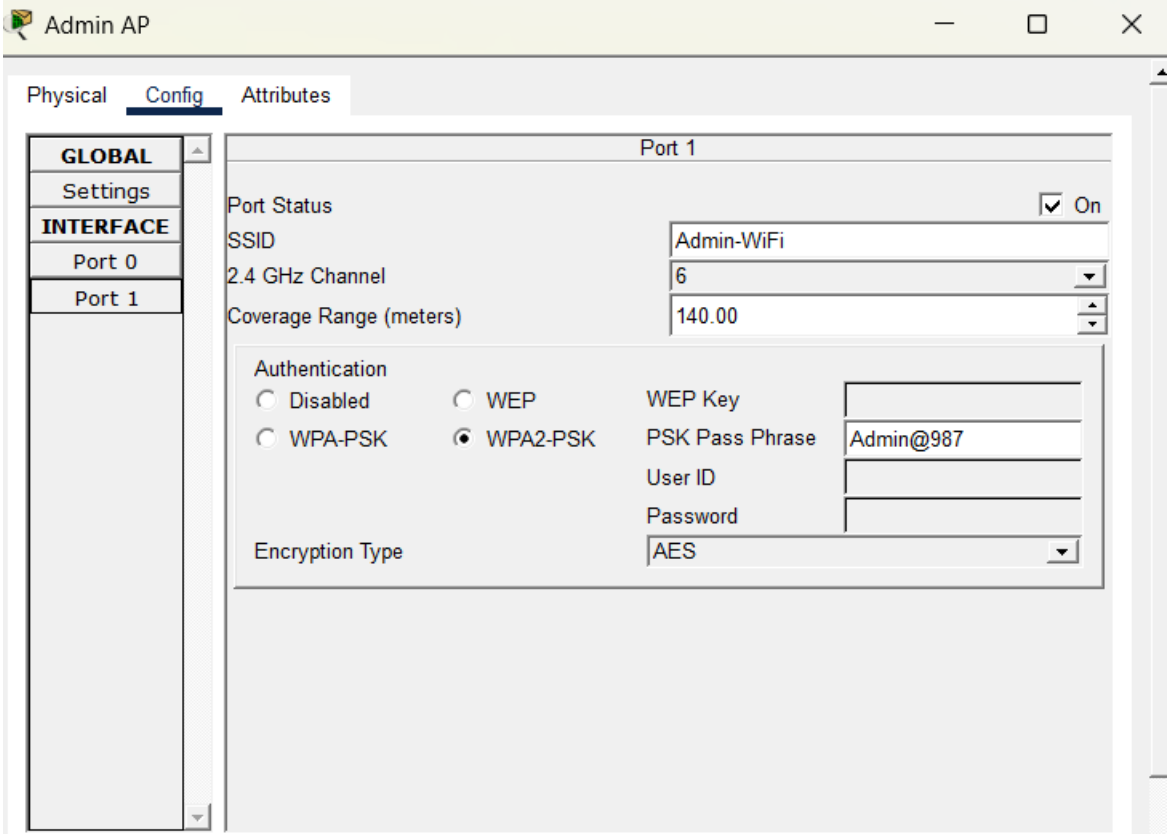
Router0

Physical Config CLI Attributes

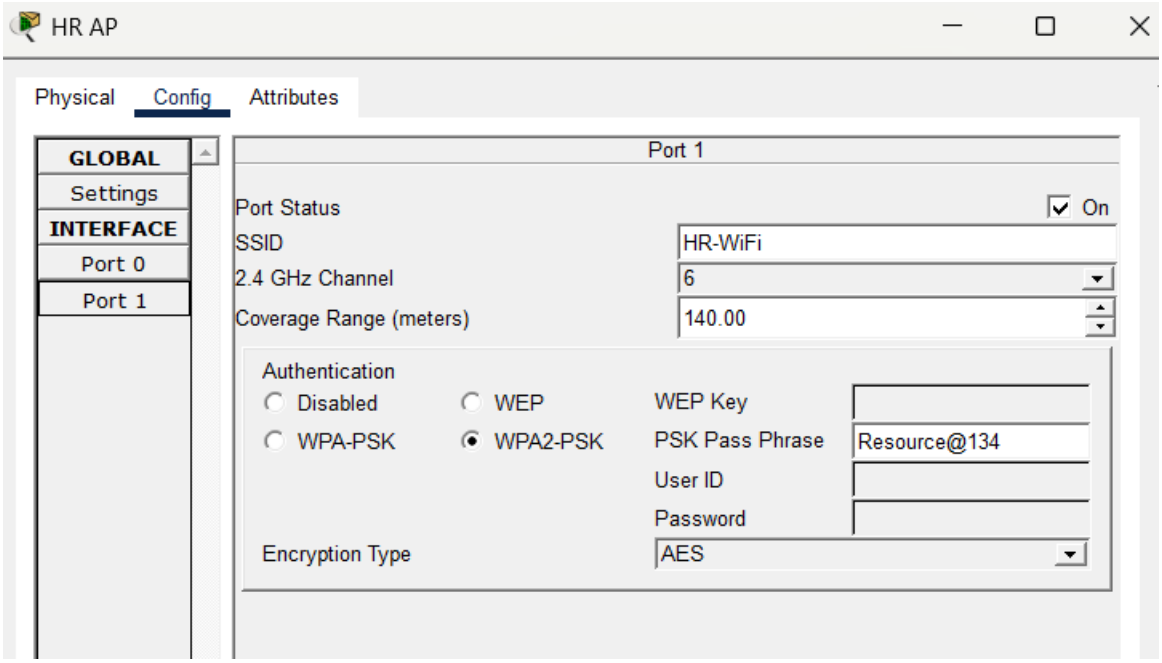
```
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp pool vlan 20
                                   ^
% Invalid input detected at '^' marker.

Router(config)#ip dhcp pool Admin-Pool
Router(dhcp-config)#no default-router
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#do wr
Router(dhcp-config)#x
                    ^
% Invalid input detected at '^' marker.

Router(dhcp-config)#ex
Router(config)#ip dhcp pool HR-Pool
Router(dhcp-config)#no default-router
Router(dhcp-config)#default-router 192.168.1.65
Router(dhcp-config)#do wr
Router(dhcp-config)#ex
Router(config)#ip dhcp pool Finance-Pool
Router(dhcp-config)#no default-router
Router(dhcp-config)#default-router 192.168.1.129
```

Wi-Fi SETTINGS ON ACCESS POINTS.

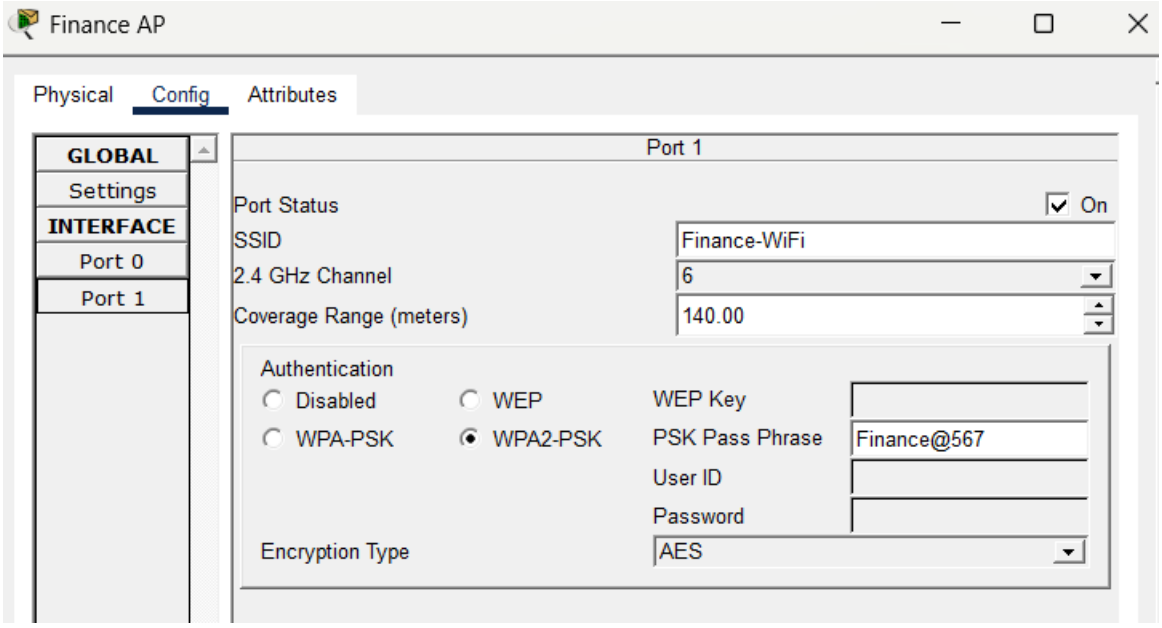
Admin Department



HR Department

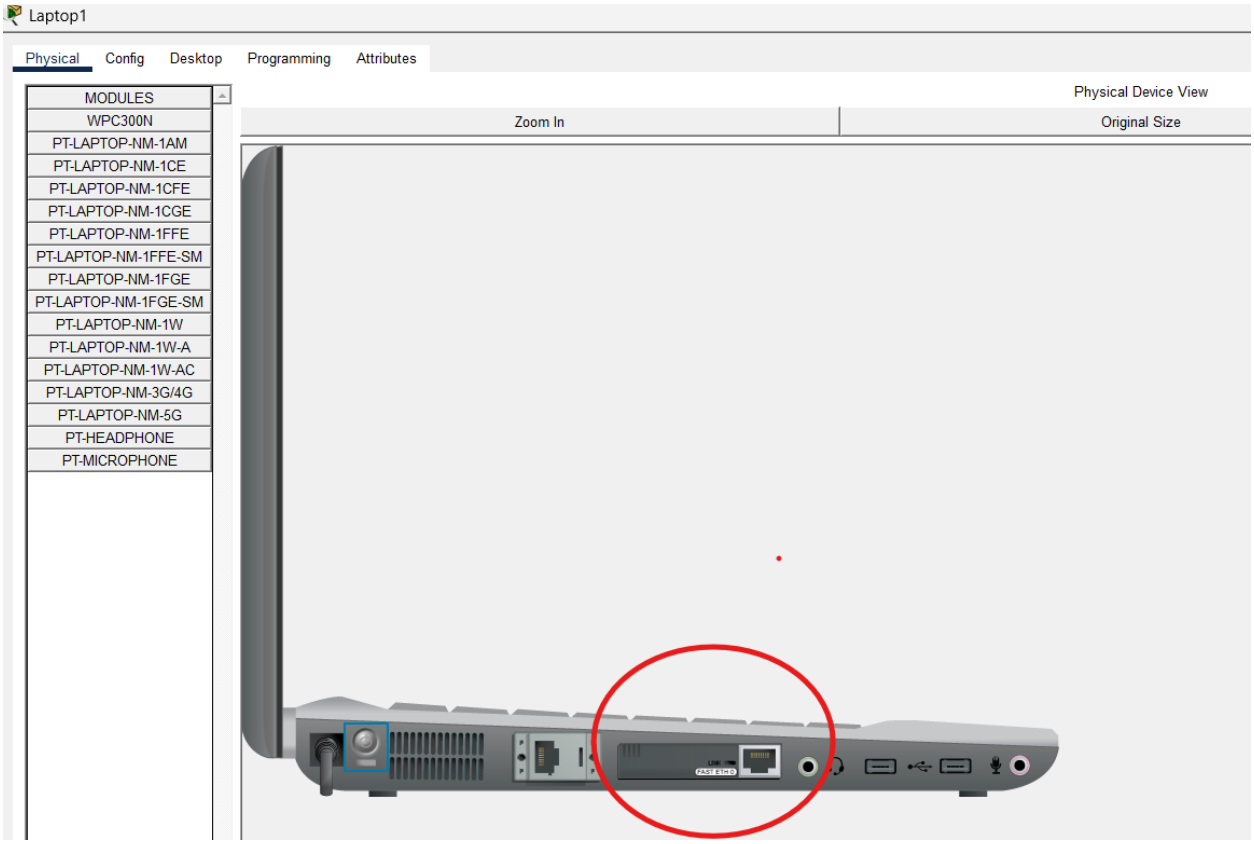


Finance Department



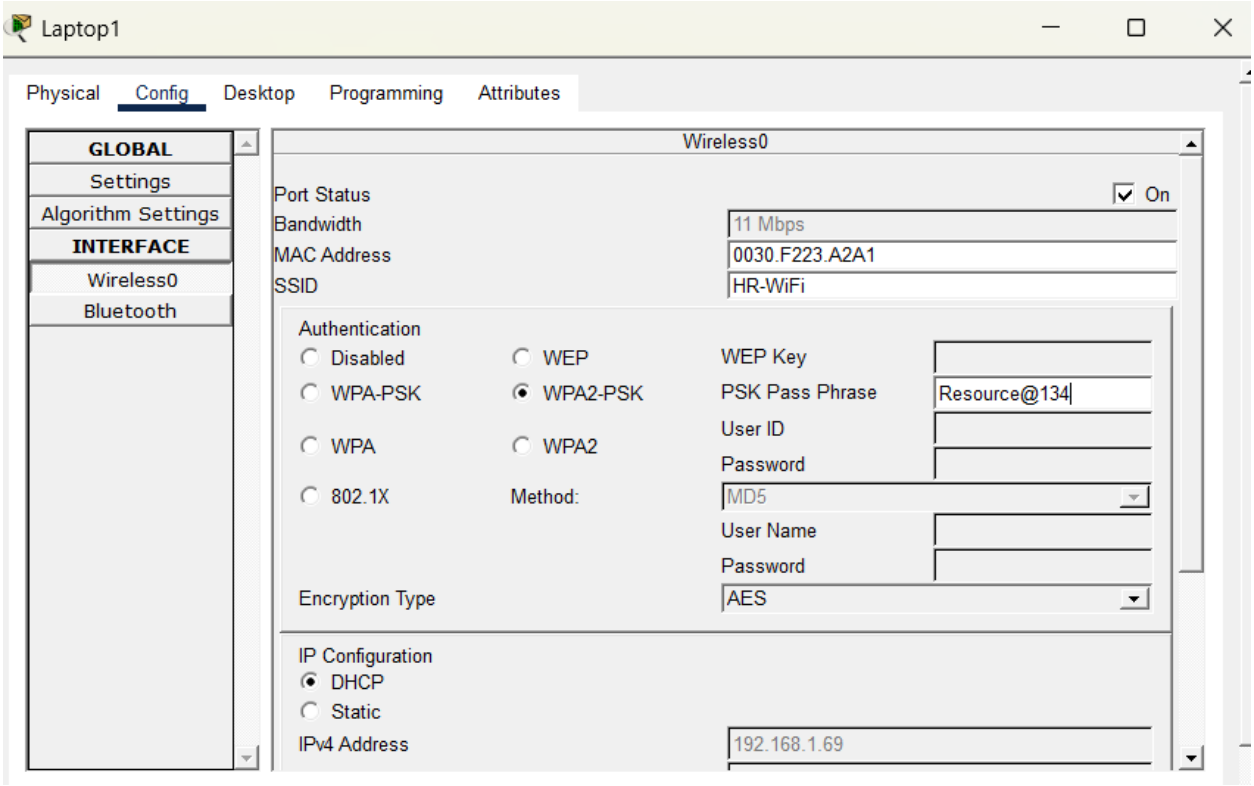
WIRELESS CONNECTIVITY SETTINGS ON END DEVICES

On Laptop



1. Turn of the laptop (Button with blue outline)
2. Removing the encircled module and replacing it with **WPC300N**
3. Then turn Laptop Back **ON**
4. Make sure **DHCP** is **Enable** in **DESKTOP** settings **IP Address.**
5. In config Bar of Laptop (end device**)**

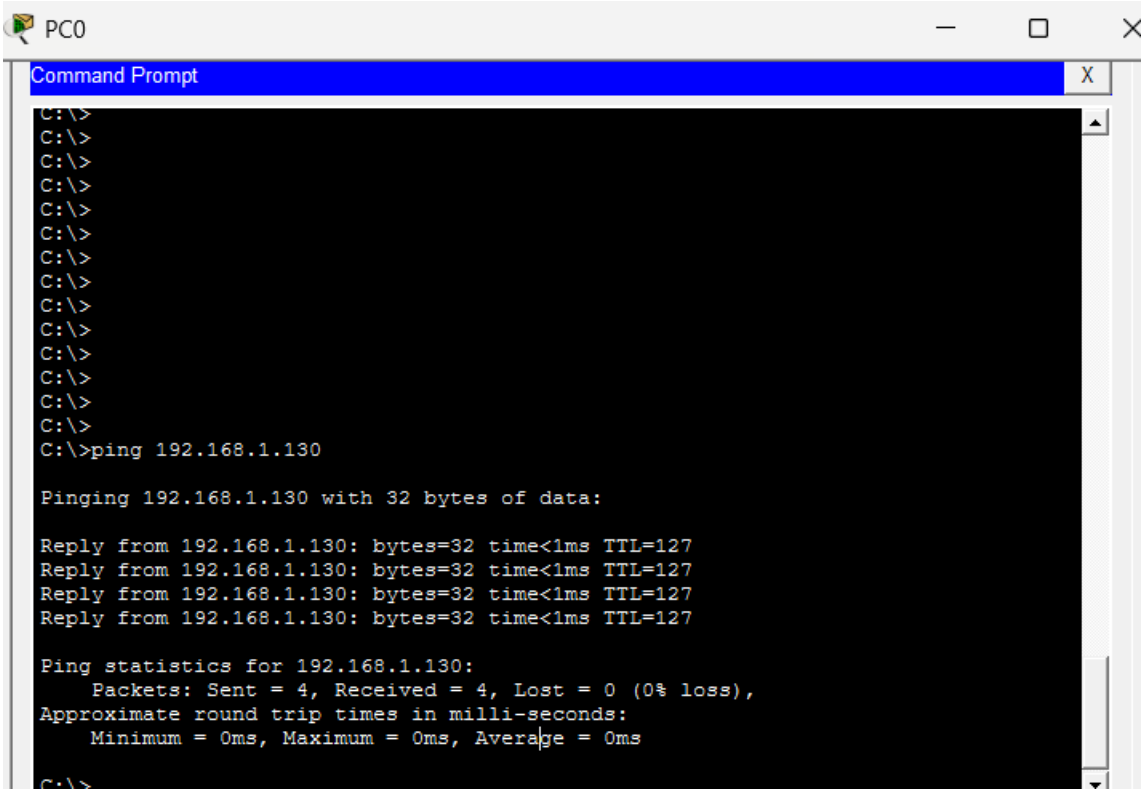
- Open Wireless Connection Module.
- Enter the SSID and Pass Key.



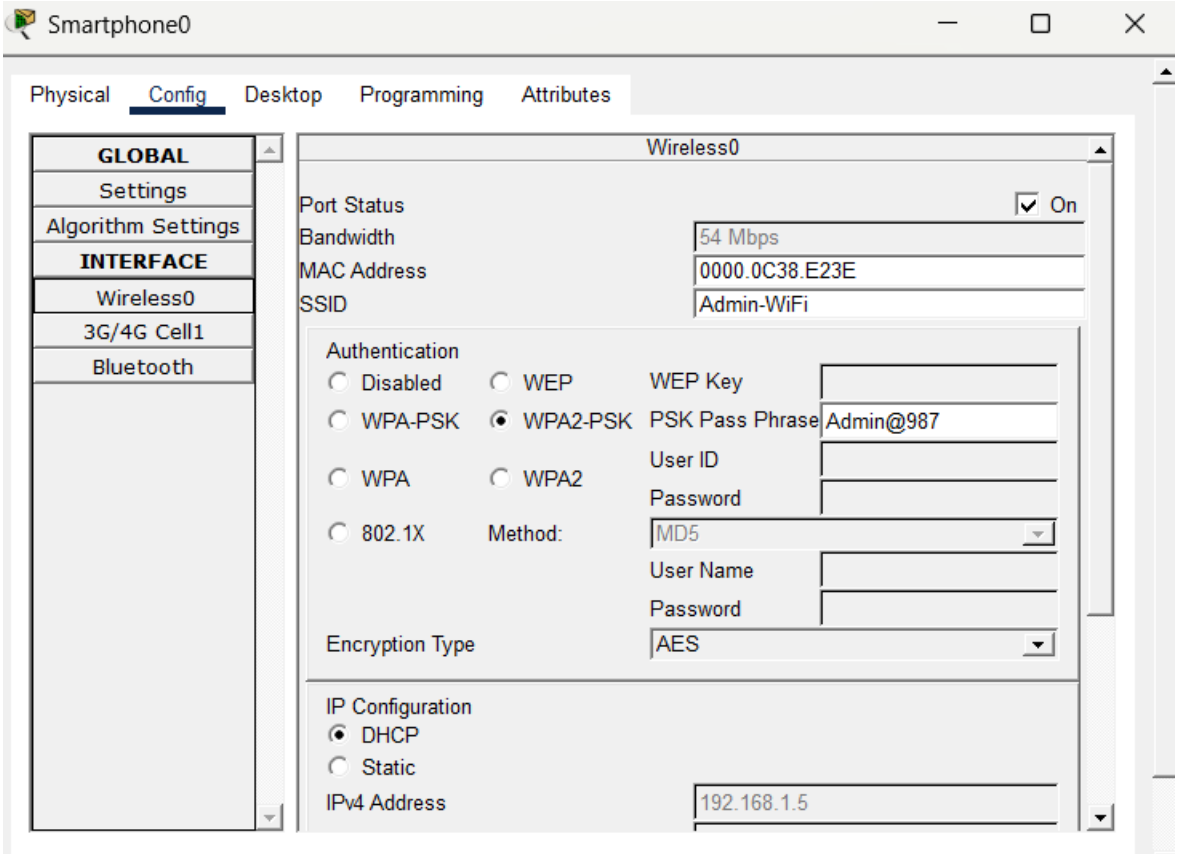
TESTING AND VERIFICATION

Inter-VLAN Ping.

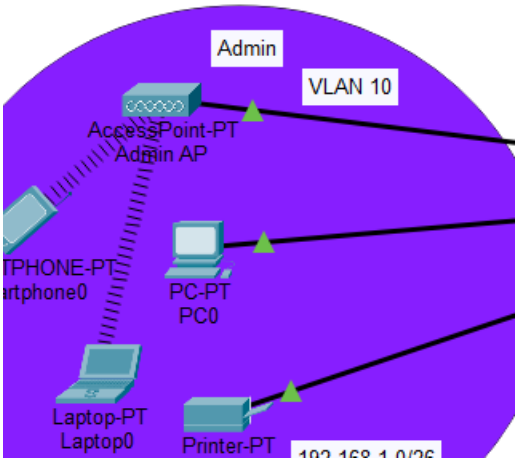
Pinging PC0 from VLAN 10 – PC2 VLAN 30



Wi-Fi Connectivity



(Result)



The dotted line shows wireless connection.

SECURITY CONSIDERATIONS

- VLAN segmentation improves security

- Use strong Wi-Fi passwords
- Disable unused ports
- Enable port security

Conclusion

This network design successfully meets all the requirements of XYZ Company's new branch. It ensures:

- Efficient communication between departments
- Secure segmentation using VLANs
- Wireless accessibility
- Automatic IP configuration

The design is scalable and can support future expansion.